Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 38)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
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8	
9	
10	
11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 . 1 Distinguish between **renewable** and **depletable** energy resources, giving an example of a resource that is both renewable and depletable.

[3 marks]

0 1 . 2 Explain why the abundance of an energy resource is not the same as the amount available for use. Give **two** examples.

[3 marks]

0 1 . 3 Describe the locational constraints of **two** named renewable energy resources.

[4 marks]

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0 2 Figure 1 lists the stages in a water treatment works supplying drinking water from a lowland river.

Figure 1

Stage	Process
1	Screens
2	Sedimentation
3	Aeration
4	Flocculation and clarification
5	Filtration through sand and gravel
6	Activated carbon
7	Sterilisation
8	pH adjustment

0 2 . 1 Explain why clay particles do not settle out in the sedimentation stage, and describe how they are removed in stage 4.

[3 marks]

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0 2 . 2 Explain the purpose of aeration (stage 3).

[2 marks]

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0 2 . 3 Explain why ozone or UV light may be used for sterilisation instead of chlorine when water comes from peaty upland sources.

[2 marks]

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0 2 . 4 Describe how each of the following contaminants is removed from water.

[3 marks]

Toxic metal ions (lead, mercury, arsenic)

Salt

Odours and pesticides

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0 3 A motor-racing circuit has residents living 500 m away. The circuit operator wants to compare two designs of acoustic barrier: an earth baffle mound and a timber acoustic fence.

0 3 . 1 Describe how you would carry out an investigation to compare the effectiveness of the two barriers.

[5 marks]

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0 3 . 2 Explain why readings should be taken with the same wind direction.

[1 mark]

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0 3 . 3 Explain the difference between the LA10 and LA90 measures of traffic noise.

[2 marks]

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0 3 . 4 Outline **two** ways in which the residents' houses could be designed to reduce the noise reaching occupants.

[2 marks]

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0 4 A student compared the invertebrates in a stream above and below a sewage outfall using kick sampling.

0 4 . 1 Describe how kick sampling is carried out so that the samples from the two sites are comparable.

[3 marks]

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0 4 . 2 Give **three** limitations of kick sampling.

[3 marks]

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0 4 . 3 Explain how a Surber sampler improves on kick sampling.

[2 marks]

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0 4 . 4 Explain why the invertebrates found give a better indication of long-term sewage pollution than a single measurement of dissolved oxygen.

[2 marks]

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0 5 A nuclear reprocessing plant on the coast discharges low-level liquid radioactive waste into the sea through a pipeline.

0 5 . 1 Explain what is meant by **Critical Pathway Analysis** (CPA).

[2 marks]

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0 5 . 2 Give **three** factors that would be considered in a CPA for this discharge.

[3 marks]

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0 5 . 3 Describe how **Critical Group Monitoring** (CGM) would be used to assess the risk to the public from this discharge.

[3 marks]

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0 5 . 4 Suggest **two** types of environmental sample that would be collected to check the CPA predictions.

[2 marks]

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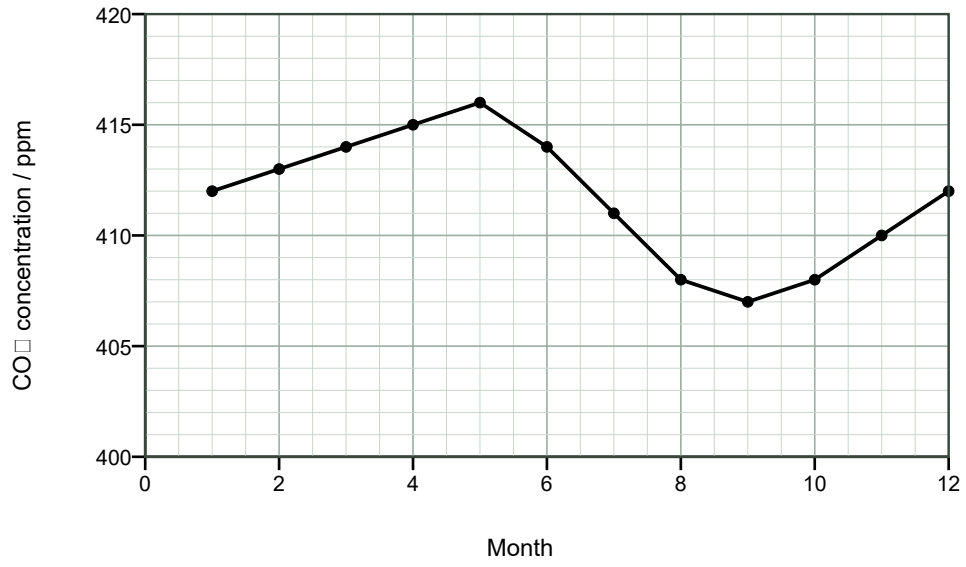
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0 6 Table 1 shows the composition of the dry atmosphere. Figure 1 shows the changes in carbon dioxide concentration measured at a site in the northern hemisphere over one year.

Table 1

Gas	Percentage by volume / %
Nitrogen	78
Oxygen	21
Argon and other rare gases	0.9
Carbon dioxide	0.04
Ozone	0.000007

Figure 1



0 6 . 1 Describe and explain the pattern shown in Figure 1.

[4 marks]

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0 6 . 2 Explain why the atmosphere is described as being in a state of **dynamic equilibrium**.

[2 marks]

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0 6 . 3 Give **two** ways in which the atmosphere supports life other than by providing gases for biological molecules.

[2 marks]

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0 6 . 4 Name the two layers of the atmosphere most affected by human activities.

[2 marks]

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0 7 Figure 1 shows the number of large oil spills (over 700 tonnes) from tankers worldwide and the amount of oil transported by sea between 1975 and 2015.

Figure 1

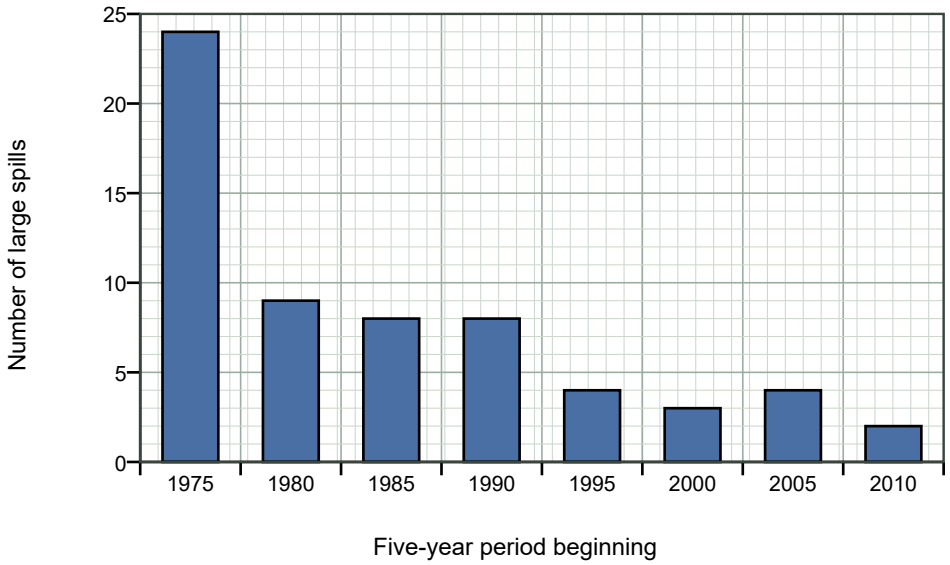


Table 1

Year	1975	1985	1995	2005	2015
Oil transported by sea / million tonnes	1 400	1 200	1 800	2 400	2 900

0 7 . 1 Calculate the percentage increase in oil transported by sea between 1975 and 2015.

[2 marks]

Answer ..... %

0 7 . 2 Describe the trend in the number of large oil spills shown in Figure 1.

[2 marks]

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0 7 . 3 Suggest **three** reasons for the decline in the number of large spills despite the increase in oil transported.

[3 marks]

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0 7 . 4 Explain why the total amount of oil pollution entering the sea may not have fallen as much as the number of large tanker spills.

[3 marks]

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0 8 . 1 Define the term **persistence** as applied to a pollutant.

[1 mark]

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0 8 . 2 Explain how the persistence of a pesticide is affected by **two** named environmental factors.

[4 marks]

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0 9 . 1 Describe **three** factors that have restricted the growth of nuclear power.

[3 marks]

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0 9 . 2 Explain how a plutonium 'breeder' reactor can produce more fissile fuel than it uses.

[3 marks]

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0 9 . 3 Give **two** advantages and **two** disadvantages of thorium reactors compared with uranium reactors.

[4 marks]

Advantages .....

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Disadvantages .....

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1 0 . 1 Evaluate the success of the methods and strategies used to reduce global climate change.

[9 marks]

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Extra space

1 0 . 2 State what is meant by a **carbon footprint**.

[1 mark]

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Write an essay on one of the following topics.

1 1 . 1 Discuss how the atmosphere supports life on Earth and evaluate the extent to which human activities threaten these life-support functions.

[25 marks]

OR

1 1 . 2 Discuss how the design of a sampling programme and the choice of statistical analysis affect the validity of conclusions drawn from environmental investigations.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS